Practical 05 — Laboratory Submission Report

Settings in a File, Bugs Caught by a Robot · SCSE3040 MLOps

Session 2026-27

1. Student and Laboratory Session Details

Name:	Parth Rawat
Enrollment / Roll No:	S24CSEU1345
Batch:	23
Laboratory Date:	October 20, 2026
Course:	SCSE3040 — Machine Learning Operations (Bennett University)

2. Repository Link (Written out in full as text)

https://github.com/Daemon-01/MLOps.git

https://github.com/Daemon-01/MLOps

3. Contents of config.yaml (Pasted as text)

```yaml
# Settings for the delivery-time model.
# Change values here, never in the code.

data:
  path: ../data/delivery_times.csv
  features:
    - distance_km
    - prep_time_min
    - traffic_level
    - rain
  target: delivery_min

split:
  test_size: 0.2
  seed: 42

model:
  name: linear_regression

training:
  max_rows: 300
```

Practical 05 — Test Results & Model Verification

Automated Testing with pytest · Refactoring Verification

10 / 10 Tests Passing

4. Screenshot of Terminal after Running pytest (Full Summary Line Visible)

PowerShell -- work -- pytest -v

```
PS C:\Users\ADMIN\Downloads\MLops\work> python -m pytest -v
============================= test session starts ==============================
platform win32 -- Python 3.11.9, pytest-9.1.1, pluggy-1.6.0 -- C:\Users\ADMIN\Downloads\MLops\.venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\ADMIN\Downloads\MLops
configfile: pyproject.toml
plugins: anyio-4.14.2
collecting ... collected 10 items

test_model.py::test_rain_never_makes_delivery_faster PASSED              [ 10%]
test_model.py::test_further_is_never_faster PASSED                       [ 20%]
test_model.py::test_error_is_better_than_guessing PASSED                 [ 30%]
test_model.py::test_heavier_traffic_is_never_faster PASSED               [ 40%]
test_orders.py::test_minutes_per_km_simple PASSED                        [ 50%]
test_orders.py::test_minutes_per_km_rejects_zero_distance PASSED         [ 60%]
test_orders.py::test_good_order_is_accepted PASSED                       [ 70%]
test_orders.py::test_bad_traffic_level_is_rejected PASSED                [ 80%]
test_speed.py::test_slow_delivery PASSED                                 [ 90%]
test_speed.py::test_negative_distance_raises PASSED                      [100%]

============================== 10 passed in 0.06s ==============================
```

5. Model Output Statement (Before vs. After Refactor)

One-line Statement: Before the refactor, the hardcoded model training produced an MAE of **1.92 minutes**, and after refactoring to use `config.yaml`, the package produces the identical honest MAE of **1.92 minutes** on the complete dataset (and **2.18 minutes** when restricted to the configured 300-row limit in Task T1).